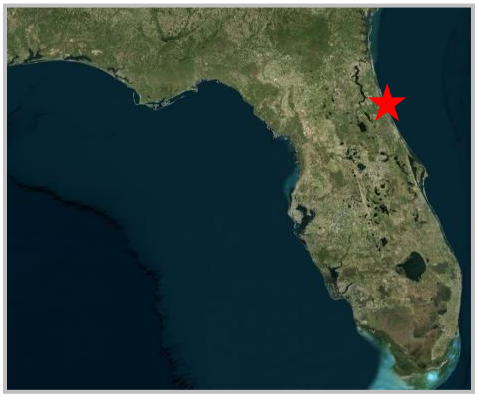
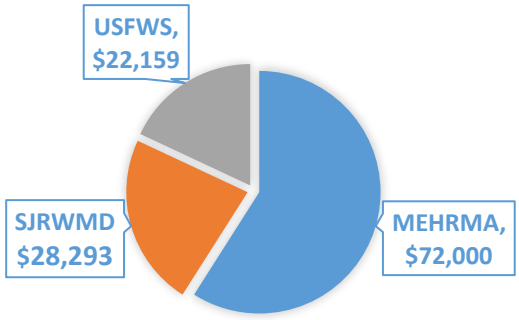
Dragline-ditched Saltmarsh Restoration, Flagler County, FL

FY 18/19



Project Partners and Funding



Project Description



Figure 1 Amphibious excavators conducting saltmarsh restoration in the Flagler project area.

The Flagler dragline-ditched saltmarsh restoration project is a result of a largescale restoration initiative which has been conducted throughout the east coast of Florida over the past two decades. FWC has partnered with regional entities (St. Johns River Water Management District, Volusia County) and a federal partner (USFWS) to conduct coastal wetland restoration, sometimes using MEHRMA funding. The USFWS provides funds through the National Coastal Wetlands Conservation Grant Program (NCWCGP). Coastal wetlands in this region were greatly altered by the practice of dragline ditching during the mid-1900s as a mosquito control measure. The process left numerous ditches with adjacent spoil piles thereby reducing the marsh footprint. The original objective of this project was to restore more than 250 acres of dragline ditch-impacted wetlands on public lands in Volusia County. Following the restoration of 103 acres in Volusia during 2013-15, the secondary objective is to restore 96 acres of dragline ditch-impacted

wetlands on public lands in Flagler County. Using the proven restoration technique, vegetation is cleared from the spoil area and placed in the adjacent ditch. This process avoids burning the plants and permanently sequesters the carbon they contain. Spoil material is moved to the side "fingers" of the ditch restoring marsh elevations or shoaling finger ditches. If additional material remains, the main ditch is narrowed. The area of the spoil pile and the newly-filled ditch are graded to the adjacent wetland elevation. This results in substantially more marsh area at coastal wetland elevation while still providing effective mosquito control.

Benefits

- Restores wetland area and function
- Increases production of wetland-dependent species including fishes and decapods
- Provides effective mosquito control

FWC Strategic Plan Elements Supported

Theme 1, Goal 2: Make sure there are sufficient quantities and quality of habitats to support healthy and diverse fish and wildlife populations.

Strategies:

- Secure and maintain sufficient interconnected quantities of habitats to sustain healthy fish and wildlife populations.

Acreage restored by habitat type (footprint)

This project entails the restoration of ~96 acres of degraded coastal wetland sites which are part of an area of wetlands and open water that is ~250 acres:

- Restored sites increase in wetland area by ~40%

Acreage affected by restoration activities: 96 acres





Figure 2: Native species such as wetland plants and fiddler crabs quickly colonize the restored areas.
Figure 3: Fishing adjacent to the project work site can be productive and the project benefits many fishery species .